Reinforcement Learning over attachment probability p

We model the control problem as a Markov decision process (MDP)

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma).$$

At each time step t, the state is

$$s_t = \left(C_t^{\max}, C_t^{\min}, C_t^{\text{avg}}, r_t^{\max}, r_t^{\min}, r_t^{\text{avg}}\right).$$

The policy chooses a continuous control action

$$a_t \sim \pi_\theta(a_t \mid s_t), \qquad a_t = p_t \in [0, 1].$$

The graph of potential lender links is sampled as an Erdős-Rényi network

$$G_t \sim \mathcal{G}(N, p_t), \qquad \Pr[(i, j) \in E_t] = p_t.$$

The attachment probability directly controls the expected graph density

$$\mathbb{E}[|E_t|] = p_t \, \frac{N(N-1)}{2}.$$

This affects the set of possible lenders selected by the model and therefore the loan matching process.
The immediate reward is

$$\mathcal{R}_t = \text{loans}_t - \lambda_b \, \text{bankruptcies}_t,$$

with $\lambda_b > 0$ a penalty weight.

The objective is to maximize the expected discounted return

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T-1} \gamma^t \mathcal{R}_t\right].$$

PPO optimizes the clipped surrogate objective

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\left[\min\left(\rho_t(\theta)\,\hat{A}_t, \text{clip}(\rho_t(\theta), 1-\varepsilon, 1+\varepsilon)\,\hat{A}_t\right)\right],$$

where

$$\rho_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}.$$

The advantage estimate is computed with GAE:

$$\delta_t = \mathcal{R}_t + \gamma V(s_{t+1}) - V(s_t), \qquad \hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}.$$